TECHNOLOGY RAG ASSISTANT

Architecture Explanation

Objective

The objective of this project is to build a Technology Support Assistant that answers user questions using only information retrieved from provided documents.

Document Ingestion

Technology-related documents are loaded from a local folder and processed using LangChain document loaders.

The documents are split into smaller semantic chunks using RecursiveCharacterTextSplitter to improve retrieval accuracy.

Embeddings

Each chunk is converted into a vector representation using the Ollama embedding model (nomic-embed-text).

These vectors capture semantic meaning and allow similarity-based retrieval.

Vector Database

The generated embeddings are stored inside a FAISS vector database.

FAISS enables efficient similarity search over document chunks.

Retrieval

When a user submits a question, the retriever searches FAISS and returns the most relevant chunks.

These retrieved chunks become the context for answer generation.

Response Generation

The retrieved context and conversation history are passed to the Mistral language model running through Ollama.

The model generates answers using only the retrieved information.

Conversation Memory

The chatbot maintains conversation history so that follow-up questions such as “Where is it used?” or “Does it support async operations?” can be understood correctly.

Hallucination Prevention

The assistant is instructed to refuse answering when relevant information cannot be found in the retrieved documents.

Response: “I don't have enough information in the provided documents.”

Workflow

User Query
↓
Retriever
↓
FAISS Search
↓
Relevant Chunks
↓
Mistral LLM
↓
Grounded Response

Conclusion

The Technology Support Assistant successfully demonstrates a complete Retrieval-Augmented Generation (RAG) pipeline using LangChain, FAISS, Ollama, and local embedding models. The system ingests technology-related documents, converts them into vector embeddings, stores them in a FAISS vector database, and retrieves relevant information to answer user queries.

The assistant supports context-aware conversations by maintaining conversation history, allowing it to handle follow-up questions effectively. In addition, the system reduces hallucinations by restricting responses to retrieved document content and refusing to answer when sufficient information is unavailable.

Overall, the project demonstrates how retrieval, vector search, and large language models can be combined to build a practical domain-specific conversational assistant that provides accurate, grounded, and context-aware responses.